

Liquid Propellant Rocket Engines

LPRE

Liquid propellants $\begin{matrix} \nearrow \text{Ox} \\ \searrow \text{Fuel} \end{matrix}$

Monopropellants

(capable of igniting or exothermic decomposition)

Ex: Hydrazine (N_2H_4)

Bipropellants

O_2 - H_2

O_2 - RPI

C+H+N

N_2O_4 - MMH [CH_3NHNH_2]

N_2O_4 - UDMH [$(\text{CH}_3)_2\text{NNH}_2$]

Desirable propellant characteristics (from Sutton, "Rocket Propulsion Elements.")

① Low freezing point

② High specific gravity ($\text{SG} = \frac{\rho}{\rho_{\text{H}_2\text{O}}}$)

$$m = \rho V \Rightarrow V = \frac{m}{\rho}$$

③ stability

④ high specific heat, high thermal conductivity, and high boiling point for propellants used for cooling

⑤ Low vapor pressure is a desirable pumping characteristic

⑥ Variation of physical properties with temperature should be small and/or similar for fuel and oxidizer.

⑦ readily ignitable

Oxidizer - fuel mass ratio (r): (aka mixture ratio)

$$\text{OF} = r = \frac{\dot{m}_{\text{ox}}}{\dot{m}_{\text{f}}} = \frac{\dot{m}_{\text{ox}} \bar{M}_{\text{ox}}}{\dot{m}_{\text{f}} \bar{M}_{\text{f}}}$$

$$\text{OF} = r = R$$

Average bulk density (ρ_{bulk})

$$\rho_{\text{bulk}} = \frac{\dot{m}_{\text{f}} + \dot{m}_{\text{ox}}}{V_{\text{f}} + V_{\text{ox}}} = \frac{m_{\text{f}} \left(1 + \frac{m_{\text{ox}}}{m_{\text{f}}}\right)}{\frac{m_{\text{f}}}{\rho_{\text{f}}} + \frac{m_{\text{ox}}}{\rho_{\text{ox}}}}$$

total mass
" Volume

$$\rho_{bulk} = \frac{(1+r)\rho_f \rho_{ox}}{\rho_{ox} + r\rho_f}$$

$O_2 + H_2$
 LOX - LH₂ rocket: $r = 4$, $SG_{LOX} = \frac{\rho_{O_2}}{\rho_{H_2O}} = 1.14$, $SG_{LH_2} = \frac{\rho_{H_2}}{\rho_{H_2O}} = 0.071$
 $\rho_{LOX} = 1140 \text{ kg/m}^3$, $\rho_{LH_2} = 71 \text{ kg/m}^3$ ($\rho_{H_2O} = 1000 \text{ kg/m}^3$)

$$\rho_{bulk} = \frac{(1+4)(71)(1140)}{1140 + (4)(71)} = 284 \text{ kg/m}^3$$

O_2 ← kerosene
 LOX - RPI rocket: $r = 2.58$, $SG_{RPI} = 0.815$
 $\rho_{RPI} = 815 \text{ kg/m}^3$

$$\rho_{bulk} = \frac{(1+2.58)(815)(1140)}{1140 + (2.58)(815)} = 1026 \text{ kg/m}^3$$

For a given propellant mass,

$$\frac{m_1 = \rho_1 V_1}{m_2 = \rho_2 V_2} \Rightarrow \frac{V_2}{V_1} = \frac{\rho_1}{\rho_2} \left(\frac{V_{\text{tank}}(LOX/LH_2)}{V_{\text{tank}}(LOX/RPI)} \right) \approx 3.6 \frac{V_{\text{tank}}(LOX/LH_2)}{V_{\text{tank}}(LOX/RPI)}$$

[Study Table 12-1]

Put $\frac{V_2}{V_1} = 1$
 $V = \frac{m}{\rho}$

$$r = \frac{\dot{m}_{ox}}{\dot{m}_f}$$

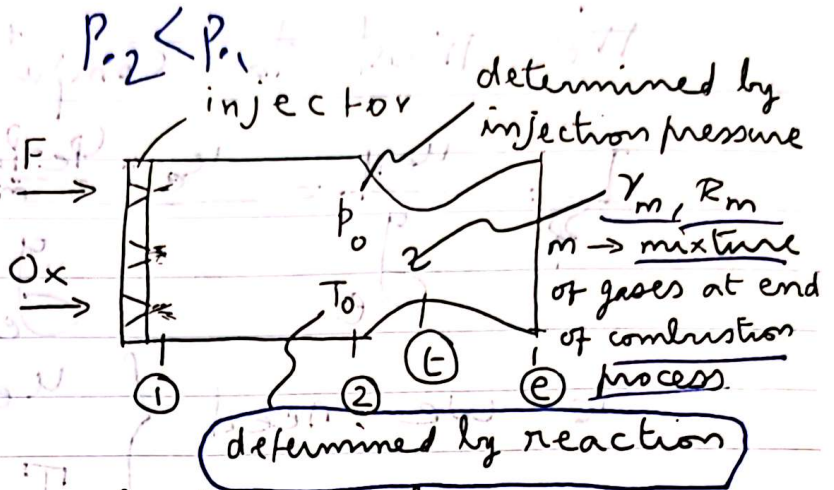
$r_2 < r_1$ means

RRP1 is greater in weight than LH₂

$$\begin{cases} m_{H_2} = \rho_{H_2} V = 71 \text{ kg} \\ m_{RPI} = 815 \text{ kg} \end{cases}$$

$$m_{RPI} \gg m_{H_2}$$

* A LPRE Thrust chamber requires propellant feed at high pressure. Unlike in the SRM, there is no increase in p_0 in the combustion chamber.



Recall: $\bar{M}_m = \sum X_i \bar{M}_i$, $\bar{c}_{p,m} = \sum X_i \bar{c}_{p,i}$, $\gamma_m = \frac{\bar{c}_{p,m}}{\bar{c}_{p,m} - R}$

For the mixture, $R_m = \frac{R}{\bar{M}_m}$

$r = \frac{\dot{m}_{O_2}}{\dot{m}_{H_2}} = 5$

Ex.



An ideal hydrogen-oxygen rocket Thrust chamber has $r = 5$. Fuel (H_2) and oxidizer (O_2) are injected at 68 atm, and 298.15 K. The temperature (T_0) and The composition (mole fractions X_i) of the product mixture are given in Table 12.2. Assuming fully-expanded jet at SL, Calculate I_{sp} . $\gamma_{MON} = 1.67$, $\gamma_{DI} = 1.29$, $\gamma_{TRI} = 1.17$

Table 12.2: Thermodynamic properties of combustion products. The table lists properties for H_2 , O_2 , H_2O , OH , H , and O at various temperatures and pressures. The table is used to find the mole fractions and specific heat capacities of the combustion products.

68 atm
298.15K

$p_0 = 68 \text{ atm} = 68(0.101325) = 6.89 \text{ MPa}$

We want to compute I_{Sp}

$$I_{Sp} = \frac{u_{eq}}{g_e} = \frac{u_e + \frac{(P_e - P_a) A_e}{\rho}}{g_e} = \frac{u_e + \frac{R T_e}{u_e} \left(1 - \frac{P_a}{P_e}\right)}{g_e}$$

$$I_{Sp} = \frac{u_{eq}}{g_e} = \frac{u_e + \frac{(p_e - p_a)A_e}{\rho}}{g_e} = \frac{u_e + \frac{RT p_e}{u_e} \left(1 - \frac{p_a}{p_e}\right)}{g_e}$$

So we need $\begin{cases} u_e \\ T_e \\ p_e \end{cases} \rightarrow$ we need M_e

M_e can be obtained from $\frac{P_{oe}}{P_e} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{\frac{\gamma}{\gamma-1}}$

This Eqⁿ needs $\begin{cases} P_e \rightarrow \text{given in question} \\ P_e \rightarrow " " " \\ X_m \rightarrow \text{needs to compute} \end{cases}$

How to obtain δ_m

1) compute $\bar{M}_m = \sum X_i \bar{M}_i = 11.59$

2) " $R_m = \frac{\bar{R}}{\bar{M}} = 717.4 \frac{\text{J}}{\text{kgK}}$

3) $\bar{c}_{p,m} = \sum \alpha_i \bar{c}_{p,i} = 5.772 \frac{\text{kg}}{\text{R}}$

4) $\gamma_m = \frac{C_{pm}}{C_{pm} - R}$

From Table 12.3,

$$T_0 = 3429 \text{ K}$$

$$x_{H_2} = 0.3607, x_{O_2} = 0.977(10^{-3}), x_{H_2O} = 0.5744,$$

$$x_{OH} = 0.0260, x_O = 0.0015, x_H = 0.0363$$

$$\{\bar{M}_H = 1, \bar{M}_O = 16\}$$

$$\bar{M}_m = \sum(x_i \bar{M}_i) = 0.3607(2) + 0.977(10^{-3})(32) + 0.5744(18) + 0.0260(17) + 0.0015(16) + 0.0363(1) = 11.59$$

$$R_m = \frac{\bar{R}}{\bar{M}} = \frac{8314.3}{11.59} = 717.4 \frac{\text{J}}{\text{kg-K}}$$

$$\bar{c}_{pm} = \sum(x_i \bar{c}_{pi})$$

$$\text{Mono} \leftarrow \bar{c}_{p_H} = \bar{c}_{p_O} = \frac{\gamma_{\text{MON}} \bar{R}}{\gamma_{\text{MON}} - 1} = 2.493 \bar{R}$$

$$\text{diatomic} \leftarrow \bar{c}_{p_{H_2}} = \bar{c}_{p_{O_2}} = \bar{c}_{p_{OH}} = \frac{\gamma_{\text{DI}} \bar{R}}{\gamma_{\text{DI}} - 1} = 4.448 \bar{R}$$

$$\text{triatomic} \leftarrow \bar{c}_{p_{H_2O}} = \frac{\gamma_{\text{TRI}} \bar{R}}{\gamma_{\text{TRI}} - 1} = 6.882 \bar{R}$$

$$\bar{c}_{pm} = \sum(x_i \bar{c}_{pi}) = 5.772 \bar{R}$$

$$\gamma_m = \frac{\bar{c}_{pm}}{\bar{c}_{pm} - \bar{R}} = \frac{5.772 \bar{R}}{5.772 \bar{R} - \bar{R}} = \frac{5.772}{4.772} = 1.21$$

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$$\bar{R}_m = \frac{\bar{R}}{\bar{M}} = \frac{8314.3}{11.59} = 717.4 \frac{\text{J}}{\text{kg-K}}$$

$$\bar{c}_{pm} = \sum(x_i \bar{c}_{pi})$$

$$\bar{c}_{p_H} = \bar{c}_{p_O} = \frac{\gamma_{MON} \bar{R}}{\gamma_{MON} - 1} = 2.493 \bar{R}$$

$$\bar{c}_{p_{H_2}} = \bar{c}_{p_{O_2}} = \bar{c}_{p_{OH}} = \frac{\gamma_{DI} \bar{R}}{\gamma_{DI} - 1} = 4.448 \bar{R}$$

$$\bar{c}_{p_{H_2O}} = \frac{\gamma_{TRI} \bar{R}}{\gamma_{TRI} - 1} = 6.882 \bar{R}$$

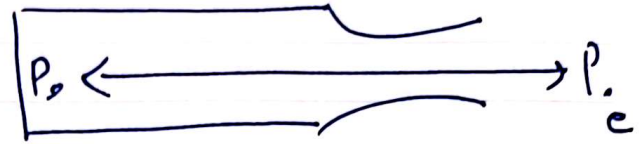
read
relevant
part in
mixture
analysis

$$\bar{c}_{pm} = \sum(x_i \bar{c}_{pi}) = 5.772 \bar{R}$$

$$\gamma_m = \frac{\bar{c}_{pm}}{\bar{c}_{pm} - \bar{R}} = \frac{5.772 \bar{R}}{5.772 \bar{R} - \bar{R}} = \frac{5.772}{4.772} = 1.21$$

Calculation of I_{sp} is as discussed earlier (ch 11)

$$p_{0e} = p_0 = 68 \text{ atm}$$



$$p_e = p_a = \underbrace{1 \text{ atm}}_{\text{sea level}} \\ \text{fully expanded}$$

$$\frac{p_{0e}}{p_e} = \frac{68}{1} = 68$$

$$\left(1 + \frac{\gamma_m - 1}{2} M_e^2\right)^{\frac{\gamma_m}{\gamma_m - 1}} = 68 \Rightarrow M_e = 3.21$$

$$T_{0e} = T_0 = 3429 \text{ K}$$

$$T_e = \frac{T_{0e}}{1 + \frac{\gamma_m - 1}{2} M_e^2} = 1647 \text{ K}$$

$$\checkmark u_e = M_e \sqrt{\gamma_m R_m T_e} = 3.21 \sqrt{(1.21)(717.4)(1647)} \\ = 3838 \text{ m/s}$$

$$\checkmark u_{eq} = u_e + \underbrace{\frac{RT_e}{u_e} \left(1 - \frac{p_a}{p_e}\right)}_0 = u_e = 3838 \text{ m/s}$$

$$\checkmark I_{sp} = \frac{u_{eq}}{g_e} = \frac{3838}{9.81} = 391 \text{ s}$$

$$\frac{A_e}{A_t} = \frac{A_e}{A_e^*} = \frac{1}{M_e} \left[\frac{2}{\gamma_m + 1} \left(1 + \frac{\gamma_m - 1}{2} M_e^2\right) \right]^{\frac{\gamma_m + 1}{2(\gamma_m - 1)}} = 8.73$$

Area-Mach
relation